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The role of microbiome-modulating supplements in managing metabolic syndrome risk factors among overweight and obese youth: a GRADE-assessed meta-analysis

Jiamin Xie^{1*}, Sho Liu¹ and Xiao Wong¹

Abstract

Background Gut microbiota modulation has been proposed as a potential intervention for managing obesity. This meta-analysis aimed to evaluate the effects of prebiotic/probiotic/synbiotic supplementation on metabolic syndrome risk factors in obese pediatrics.

Methods A comprehensive search was conducted in databases up to January 2025. Randomized controlled trials (RCTs) evaluating prebiotics/probiotics/synbiotics in children and adolescents with overweight/obesity were included. The outcomes were body weight (BW), body mass index (BMI), BMI-z score, fasting blood sugar (FBS), homeostatic model assessment for insulin resistance (HOMA-IR), insulin, total cholesterol (TC), triglycerides (TG), low-density lipoprotein-cholesterol (LDL-C), and high-density lipoprotein-cholesterol (HDL-C). Data were pooled using a random-effects model.

Results Prebiotic supplementation was associated with significant reductions in weight (SMD = -0.81; 95% CI: -1.44 to -0.19) and BMI (SMD = -0.76; 95% CI: -1.38 to -0.14), whereas BMI z-scores remained unchanged ($p > 0.05$). Probiotics and synbiotics did not significantly affect weight, BMI, or BMI z-scores ($p > 0.05$). Glycemic and lipid profile parameters were not significantly altered by any biotic supplementation ($p > 0.05$). Subgroup analyses by intervention type, duration, sample size, or baseline BMI did not reveal consistent effects ($p > 0.05$).

Conclusion Biotic supplementation has not been shown to consistently improve metabolic syndrome risk factors in overweight and obese children, except for a modest beneficial effect of prebiotics on weight and BMI (with very-low certainty of evidence); however, alternative probiotic organisms or formulations not tested to date may have different effects.

Keywords Probiotics, Prebiotics, Synbiotics, Obesity, Pediatrics

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Introduction

Childhood obesity is a major global public health concern. In China, the prevalence of obesity between 2015 and 2019 was reported at 7.9% among children and 3.6% among adolescents [1]. According to the World Health Organization (WHO), the number of overweight or obese individuals aged 5–19 worldwide reached nearly 390 million in 2016 (<https://www.who.int/news-room/fact-sheets/detail/obesity-and-overweight>). Similarly, about one-third of children in Europe were overweight or obese [2]. In low- and middle-income countries, the dual burden of rising obesity among children under five and school-aged adolescents, alongside persistent undernutrition, poses an additional public health challenge [3, 4].

The gut microbiota has recently been recognized as a pivotal environmental determinant in the progression of obesity and its associated metabolic disturbances. Conventional strategies, including lifestyle modifications and calorie restriction diets, remain the cornerstone of weight management, and emerging evidence suggests that these approaches may also influence the gastrointestinal microbiome [5–9]. Data from animal experiments and meta-analyses of human studies generally support the beneficial role of probiotics and synbiotics in modulating gut microbiota to promote weight control and improve metabolic outcomes. In contrast, evidence from clinical trials on prebiotics has been less consistent [10]. The gut microbiota contributes to the regulation of energy homeostasis and the secretion of appetite-suppressing hormones, making its modulation a promising target for obesity prevention [11–14]. A growing body of evidence underscores the critical role of gut dysbiosis in the development of metabolic disorders, such as obesity and type 2 diabetes [15, 16].

The FAO/WHO defines probiotics as live microorganisms that confer health benefits when consumed in adequate amounts, while prebiotics are fermentable dietary fibers selectively utilized by host microorganisms to promote probiotic growth. Synbiotics refer to supplements combining both probiotics and prebiotics [17, 18]. Previous studies have suggested that probiotics, prebiotics, and synbiotics may positively influence body weight and body mass index (BMI) in overweight and obese individuals [19], with potential additional benefits for glucose regulation [20] and lipid metabolism, thereby supporting cardiovascular health [21]. Existing evidence on effect of biotic on anthropometric indices, lipid and glycemic profile, in children and adolescence with overweight and obesity have shown conflicting results [22–24]. To address this gap, we conducted the first GRADE-based systematic review and meta-analysis evaluating the impact of probiotic, prebiotic, and synbiotic supplementation on anthropometric indices, lipid levels, and

glycemic markers in both adolescents and children living with overweight/obesity.

Methods

Search strategy

Computerized searches were conducted on PubMed, Web of Science, and Scopus from inception to July 2025. The search strategy included terms related to microbiome-modulating interventions the target population, and the study design, such as ('Probiotic,' 'Lactobacillus,' 'Lactocaseibacillus,' 'Limosilactobacillus,' 'Latilactobacillus,' 'Bifidobacterium,' 'Enterococcus,' 'Prebiotic,' 'Oligosaccharides,' 'Galacto-oligosaccharide,' 'Fructooligosaccharide,' 'Synbiotic') combined with terms related to obesity and overweight ('Obesity,' 'Overweight,' 'Adiposity,' 'Adipose Tissue Hyperplasia') and the population ('Child,' 'Adolescent,' 'Teen,' 'Youth,' 'Juvenile,' 'Minors'), as well as study design filters ('Randomized Controlled Trial,' 'Placebo,' 'Blind,' 'Controlled Trial'). The specific strategy of each database is provided in **Supplementary materials**. No language and date restrictions were applied to maximize the inclusion of eligible studies. In addition to peer-reviewed publications, we included grey literature (e.g., conference abstracts, theses, and trial registries) to minimize the publication bias.

Inclusion and exclusion criteria

Two reviewers independently screened RCTs for eligibility based on the predefined inclusion and exclusion criteria. Reviewer disagreements were resolved through discussion, with a third reviewer consulted when necessary. The population of interest was children and adolescents aged 2–20 years with overweight or obesity, defined as BMI-for-age and sex at or above the 85th percentile. Eligible interventions were supplementation with prebiotics, probiotics, or synbiotics, and we only included probiotic trials that clearly identified the organisms or strains administered. Studies that described interventions solely in generic terms, such as “fermented milk” without specification of microbial content, were excluded. Comparators included placebo, usual care, or no intervention. The outcomes were body weight (BW), BMI, BMI-z score, fasting blood sugar (FBS), homeostatic model assessment for insulin resistance (HOMA-IR), insulin, total cholesterol (TC), triglycerides (TG), low- and high-density lipoprotein-cholesterol (LDL-C and HDL-C), and high-density lipoprotein-cholesterol (HDL-C). Only RCTs that reported mean changes with standard deviations or provided sufficient data to calculate effect sizes were included.

Studies were excluded if full-text data were unavailable, if they were published in non-English languages, if relevant outcomes were missing, if no control group was included, or if they were duplicates. Observational

studies, case reports, and other non-randomized designs, as well as trials conducted exclusively in adult populations, were also excluded.

Screenings and data extraction

Study selection and data extraction were performed independently by two reviewers. Any disagreements were discussed and resolved with input from a third investigator. A standardized data collection form was used to extract information from each eligible study, including: first author, year of publication, country, study design, sample size, age of participants, duration and dosage of probiotic/prebiotic/synbiotic supplementation, follow-up period, type of intervention, and reported outcomes. Extracted outcomes encompassed anthropometric indices (weight, BMI, BMI-z score, FBG, insulin, HOMA-IR, TC, TG, HDL-C, and LDL-C).

Two reviewers independently screened titles/abstracts and full texts. Inter-rater reliability was assessed using Cohen's kappa (κ), with discrepancies resolved by discussion or adjudication by a third reviewer. The agreement between reviewers was substantial, with $\kappa = 0.81$ for title/abstract screening and $\kappa = 0.84$ for full-text screening.

Risk of bias assessment and meta-evidence

We evaluate the quality of the included RCTs using Version 2 of the Cochrane Risk-of-Bias tool (RoB 2) [25]. This tool looks at five domains where bias might occur: the randomization process, deviations from intended interventions, missing outcome data, outcome measurement, and selection of reported results. After reviewing these domains, each study was given an overall rating of 'low risk', 'some concerns', or 'high risk' of bias [26]. Discrepancies were resolved through discussion or adjudication by a third reviewer.

The overall certainty of evidence for each outcome across RCTs was assessed using the GRADE framework [27]. Evidence from randomized controlled trials began as high certainty and was downgraded by one or two levels if concerns were identified regarding risk of bias, inconsistency, indirectness, imprecision, or publication bias. Two reviewers conducted the GRADE assessments independently, and disagreements were resolved through discussion.

Statistical analysis

The meta-analysis was conducted using STATA version 16 (Stata Corp LP, College Station, TX, USA). Effect sizes were expressed as standardized mean differences (Hedges' g) with 95% confidence intervals. Hedges' g was chosen over Cohen's d to account for small sample sizes in several included trials [28], as it provides an unbiased estimate of standardized effect sizes in meta-analyses with limited study populations. Effect sizes were

calculated using the mean changes and standard deviations (SDs) of lipid profiles, obesity, and glycemic indices from baseline to the final follow-up in both intervention and control groups. When trials did not report changes in mean and SD, we derived change scores by subtracting baseline from final values, and calculated SDs using the formula recommended in the Cochrane Handbook, assuming a correlation coefficient of 0.5 between baseline and follow-up values [29]. Between-study heterogeneity was assessed using the I^2 index, with values greater than 50% considered indicative of substantial heterogeneity. Subgroup analyses were pre-specified in the study protocol and included intervention type, study duration, sample size, and BMI to identify potential sources of heterogeneity. To examine the robustness of the results, sensitivity analyses were carried out by removing one study at a time and observing its effect on the overall effect size. Publication bias was assessed using funnel plots only when at least 10 studies were available for an outcome, as well as Egger's regression test and Begg's rank correlation test. Where evidence of publication bias was suggested, the trim-and-fill method was applied to estimate the number of potentially missing studies and to provide adjusted pooled effect sizes. A p -value < 0.05 was deemed statistically significant for all analyses.

Results

Study selection

The initial database search yielded 1218 records. After eliminating 279 duplicates, 939 unique articles remained for screening by title and abstract. Of these, 911 were excluded because they did not align with the predefined objectives, population, or outcomes. Full texts of 28 potentially relevant articles were reviewed in detail, leading to the exclusion of 11 studies. In total, 17 studies fulfilled the eligibility criteria and were included in the final meta-analysis (Fig. 1).

Study characteristics

Among the included RCTs, weight was reported in ten RCTs [22, 24, 30–37], BMI in eight RCTs [22–24, 30, 32, 35–37], BMI-z score in ten RCTs [22, 31, 33–35, 37–41]. Glycemic outcomes were less consistently assessed: FBS was reported in eight RCTs [24, 31, 35, 37, 39–42], HOMA-IR in four RCTs [31, 35, 37, 39], insulin in five RCTs [31, 35, 37, 39, 43], and Lipid profile parameters, including TC, LDL-C, HDL-C, and TG, were reported in seven studies [31, 32, 35, 39–42]. Six studies investigated probiotics [22–24, 31, 37, 39], six evaluated synbiotics [30, 32–35, 42], and five examined prebiotics [36, 38, 40, 41, 43]. The included trials were published between 2012 and 2023, with participant ages ranging from 6 to 18 years and sample sizes varying from 9 to 103 participants. Intervention durations varied between 4 and 24

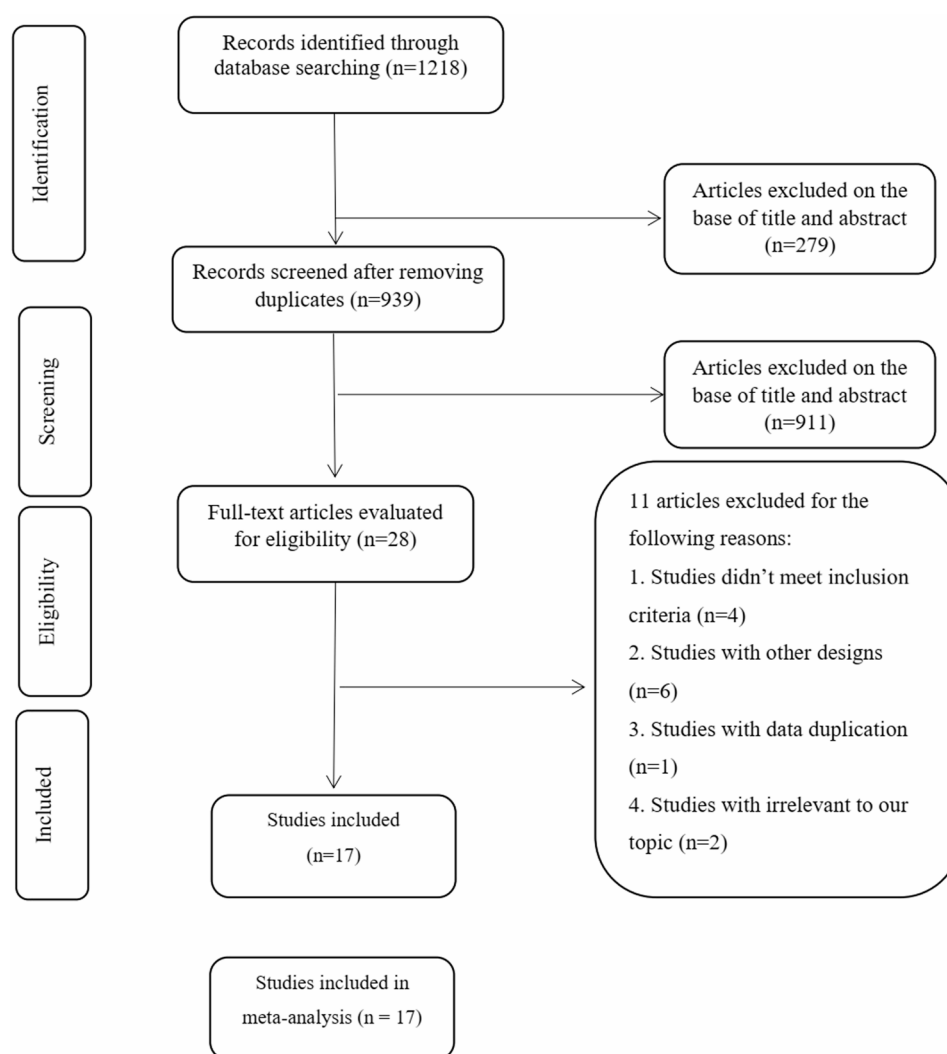


Fig. 1 PRISMA flow diagram. Articles could be excluded for more than one reason; therefore, the total number of reasons for exclusion exceeds the number of excluded records

weeks across the included trials. Study characteristics are presented in Table 1.

Risk of bias assessment and meta-evidence

Some RCTs had concerns related to outcome measurement or reporting, and three were rated as high overall risk [22, 34, 36]. The methodological quality of the eligible RCTs is detailed in Table 2. The quality of evidence was generally very low for obesity indices (weight, BMI, BMI-z) and low for glycemic and lipid profiles (FBS, HOMA-IR, insulin, TC, LDL-C, HDL-C, and TG) (Table 3).

Biotic supplementation and obesity indices

The pooled analysis demonstrated a significant weight (SMD = -0.81; 95% CI: -1.44 to -0.19; $p=0.047$) (Fig. 2A), and BMI reduction (SMD = -0.76; 95% CI: -1.38 to -0.14; $p=0.047$) (Fig. 2B), but not BMI-z scores

(SMD = 0.06; 95% CI: -0.63 to 0.74; $p=0.079$) (Fig. 2C) following prebiotics supplementation. The pooled analysis demonstrated that probiotics and synbiotics supplementation did not significantly reduce weight, BMI, and BMI z-scores (Fig. 2A-C). Overall heterogeneity was substantial for weight ($Q=64.93$, $p<0.001$; $I^2=90.0\%$), BMI ($Q=23.26$, $p=0.002$; $I^2=70.67\%$), and BMI z-score ($Q=19.69$, $p=0.020$; $I^2=55.62\%$). Subgroup analyses indicated that intervention type, study duration, sample size, and BMI contributed to heterogeneity. Obesity indices remained unchanged across all examined subgroups. No significant effects were observed according to BMI classification, other intervention types, duration of supplementation, or sample size, with all p -values >0.05 (Supplementary File).

Table 1 Baseline characteristics of included studies

Author, Country, Year	Study design	Sample size	Average age (years)	Baseline BMI (kg/m ²)	Health population	Intervention/Control	Duration (week)	Outcomes
Gøbel, Denmark, 2012 [31]	RCT, DB	50	13.13	NR	Obese adolescents	<i>L. salivarius</i> 10 ¹⁰ CFU/Placebo	12	Weight, BMI-z score, FBS, Insulin, HOMA-IR, TC, TG, LDL-C, HDL-C
Sanchis-Chordà, Spain, 2018 [39]	RCT, DB	48	12.54	NR	Obese children with insulin resistance	<i>B. pseudocatenulatum</i> , 10 ⁹ and 10 ¹⁰ CFU/Placebo	13	BMI-z score, FBS, Insulin, HOMA-IR, TC, TG, LDL-C, HDL-C
Marcelo, Brazil, 2022 [22]	RCT, DB	44	10.95	28.9	Obese children	<i>L. rhamnosus</i> 1.2 × 10 ⁸ CFU/Placebo	24	Weight, BMI, BMI-z score
Jones, USA, 2018 [23]	RCT, DB	19	14.67	30.7	Obese children	VSL#3, <i>S. thermophilus</i> , <i>B. breve</i> , <i>B. longum</i> , <i>B. infantis</i> , <i>L. acidophilus</i> , <i>L. plantarum</i> , <i>L. paracasei</i> , and <i>L. delbrueckii</i> /Placebo	16	BMI
Verma, USA, 2021 [37]	RCT, DB	9	15.86	45.7	Severe obesity adolescents	<i>L. plantarum</i> , <i>L. plantarum</i> , <i>L. salivarius</i> , <i>L. paracasei</i> , <i>L. delbrueckii</i> , <i>P. pentosaceus</i> , <i>B. animalis</i> , <i>B. breve</i> , 9 × 10 ¹¹ CFU/Placebo	12	Weight, BMI, BMI-z score, FBS, Insulin, HOMA-IR
Chen, China, 2022 [24]	RCT, DB	53	6–18	29.7	Overweight or obese children	<i>L. salivarius</i> (10 ⁹ CFU), <i>L. rhamnosus</i> (10 ⁹ CFU), <i>B. animalis</i> (8 × 10 ⁹ CFU)/Placebo	12	Weight, BMI, FBS
Safavi, Iran, 2013 [42]	RCT, TB	56	10.43	NR	Overweight or obese children	<i>L. Casei</i> , <i>L. Rhamnosus</i> , <i>L. Acidophilus</i> , <i>L. Bulgaricus</i> , <i>S. Thermophilus</i> , <i>B. Breve</i> , <i>B. Longum</i> 2 × 10 ⁸ CFU plus FOS, vitamin E, vitamin A, and vitamin C/Placebo	8	FBS, TC, TG, LDL-C, HDL-C
Kelishadi, Iran, 2014 [33]	RCT, TB	56	10.43	NR	Overweight or obese children	<i>L. Casei</i> , <i>L. Rhamnosus</i> , <i>S. Thermophilus</i> , <i>B. Breve</i> , <i>L. Acidophilus</i> , <i>B. Longum</i> and <i>L. Bulgaricus</i> 2 × 10 ⁸ CFU plus FOS, vitamin E, vitamin A, and vitamin C/Placebo	8	Weight, BMI-z score
Ipar, Turkey, 2015 [32]	RCT	77	12.33	27.1	Obese children	<i>L. acidophilus</i> , <i>L. rhamnosus</i> , <i>B. bifidum</i> , <i>B. longum</i> , <i>E. faecium</i> 4.3 × 10 ⁸ CFU plus FOS (625 Mg) and lactulose (400 mg), and vitamin A (6 mg), B1 (1.8 mg), B2 (1.6 mg), B6 (2.4 mg), E (30 mg), C (75 mg)/Control	4	Weight, BMI, TC, TG, LDL-C, HDL-C
Kianfar, Iran, 2018 [34]	RCT, DB	33	10.5	NR	Obese children	<i>L. casei</i> , <i>L. rhamnosus</i> , <i>S. thermophilus</i> , <i>B. breve</i> , <i>L. acidophilus</i> , <i>B. infantis</i> , and <i>L. bulgaricus</i> 10 ⁸ CFU plus FOS, vitamin E, vitamin A, and vitamin C/Placebo	12	Weight, BMI-z score
Yildirim, Turkey, 2022 [35]	RCT, DB	61	8–17	28.2	Exogenous obesity children	<i>L. acidophilus</i> (4.3 × 10 ⁸ CFU), <i>L. rhamnosus</i> (4.3 × 10 ⁸ CFU), <i>B. bifidum</i> (4.3 × 10 ⁸ CFU), <i>B. longum</i> (4.3 × 10 ⁸ CFU), <i>E. faecium</i> (8.2 × 10 ⁸ CFU), plus FOS (625 mg) and lactulose (400 mg), and vitamin A (6 mg), B1 (1.8 mg), B2 (1.6 mg), B6 (2.4 mg), E (30 mg), C (75 mg)/Placebo	12	Weight, BMI, BMI-z score, FBS, Insulin, HOMA-IR, TC, TG, LDL-C, HDL-C
Atazadegan, Iran, 2023 [30]	RCT, DB	60	11–12	24.7	Overweight or obese children	<i>L. coagulans</i> (6 × 10 ⁹), <i>L. indicus</i> (6 × 10 ⁹) plus FOS (38.5 g)/Placebo	8	Weight, BMI
Liber, Poland, 2014 [38]	RCT, DB	51	12.35	28.1	Overweight or obese children	Prebiotics (Oligofructose) (8 g for children aged 7–11 years and 15 g for children aged 12–18 years)/Placebo	24	BMI-z score
Nicolucci, Canada, 2017 [36]	RCT, DB	41	10.3	26.3	Overweight or obese children	Prebiotics (Oligofructose-enriched inulin) (8 g)/Placebo	16	Weight, BMI

Table 1 (continued)

Author, Country, Year	Study design	Sample size	Average age (years)	Baseline BMI (kg/m ²)	Health population	Intervention/Control	Duration (week)	Outcomes
Hume, Canada, 2017 [43]	RCT, DB	22	10.3	26.7	Overweight or obese children	Prebiotics (Oligofructose-enriched inulin) (8 g)/Placebo	16	Insulin
Zalewski, Poland, 2019 [41]	RCT, DB	96	10.23	NR	Overweight or obese children	Prebiotics (Glucomannan) (3 g)/Placebo	24	BMI-z score, FBS, TC, TG, LDL-C, HDL-C
Visuthranukul, Thailand, 2022 [40]	RCT, DB	103	10.5	28.3	Obese children	Prebiotics (Inulin) (13 g)/Placebo	24	BMI-z score, FBS, TC, TG, LDL-C, HDL-C

* All included studies were conducted in both-sex populations

Abbreviations: L: *Lactobacillus*; B: *Bifidobacterium*; S: *Streptococcus*; E: *Enterococcus*; NR not reported, RCT, DB double-blind randomized controlled trial, RCT, TB triple-blind randomized controlled trial, BMI body mass index, FBS fasting blood sugar, HOMA-IR homeostatic model assessment for insulin resistance, TC total cholesterol, TG triglyceride, HDL-C high density lipoprotein-cholesterol, LDL-C low density lipoprotein-cholesterol

Table 2 Results of risk of bias assessment

Author, Year	Randomization Process	Deviations from Intended Interventions	Missing Outcome Data	Measurement of Outcome	Selection of Reported Result	Overall Risk
Gøbel, Denmark, 2012 [31]	Low	Low	Low	Low	Low	Low
Safavi, Iran, 2013 [42]	Some concerns	Low	Low	Unclear	Low	Some concerns
Liber, Poland, 2014 [38]	Low	Low	Low	Low	Low	Low
Kelishadi, Iran, 2014 [33]	Low	Low	Low	Unclear	Low	Some concerns
Ipar, Turkey, 2015 [32]	Some concerns	Low	Low	Low	Low	Some concerns
Nicolucci, Canada, 2017 [36]	Low	Low	Low	Low	High	High
Hume, Canada, 2017 [43]	Some concerns	Low	Low	Low	Low	Some concerns
Jones, USA, 2018 [23]	Low	Low	Low	Unclear	Low	Some concerns
Kianfar, Iran, 2018 [34]	Some concerns	Low	Low	High	Low	High
Sanchis-Chordà, Spain, 2018 [39]	Some concerns	Low	Low	Low	Low	Some concerns
Zalewski, Poland, 2019 [41]	Low	Unclear	Low	Low	Low	Some concerns
Verma, USA, 2021 [37]	Low	Low	Low	Low	Low	Low
Yildirim, Turkey, 2022 [35]	Low	Low	Low	High	Low	Some concerns
Chen, China, 2022 [24]	Low	Low	Low	Unclear	Low	Some concerns
Marcelo, Brazil, 2022 [22]	High	Low	Low	Unclear	Some concerns	High
Visuthranukul, Thailand, 2022 [40]	Low	Unclear	Low	Low	Low	Some concerns
Atazadegan, Iran, 2023 [30]	Low	Low	Low	Low	Low	Low

Abbreviations: Low low risk of bias, High high risk of bias, Some concerns some concerns of bias, Unclear insufficient information to assess risk of bias

Table 3 Results of GRADE approach

	No of trials (Patients)	SMD (95% CI)	Risk of bias	Inconsistency	Indirectness	Imprecision	Publication bias	Quality of evidence
Weight (kg)	10 (484)	0.08 (−0.61, 0.77)	S	S	NS	S	NS	Very Low
BMI (kg/m ²)	8 (364)	−0.19 (−0.68, 0.29)	S	S	NS	S	NS	Very Low
BMI-z score (unit)	10 (551)	0.00 (−0.29, 0.29)	S	S	NS	S	NS	Very Low
FBS (mg/dl)	8 (476)	−0.12 (−0.47, 0.23)	S	NS	NS	S	NS	Low
HOMA-IR (unit)	4 (168)	−0.04 (−0.38, 0.46)	S	NS	NS	S	NS	Low
Insulin (μU/mL)	5 (190)	−0.02 (0.26, 0.3)	S	NS	NS	S	NS	Low
TC (mg/dl)	7 (491)	0.04 (−0.22, 0.3)	S	NS	NS	S	NS	Low
LDL-C (mg/dl)	7 (491)	0.00 (−0.26, 0.27)	S	NS	NS	S	NS	Low
HDL-C (mg/dl)	7 (491)	−0.14 (−0.54, 0.26)	S	NS	NS	S	NS	Low
TG (mg/dl)	7 (491)	0.00 (−0.19, 0.19)	S	NS	NS	S	NS	Low

Abbreviations: SMD standardized mean difference, CI confidence interval, FBS fasting blood sugar, HOMA-IR homeostatic model assessment for insulin resistance, TC total cholesterol, LDL-C low-density lipoprotein cholesterol, HDL-C high-density lipoprotein cholesterol, TG triglycerides, S serious, NS not serious

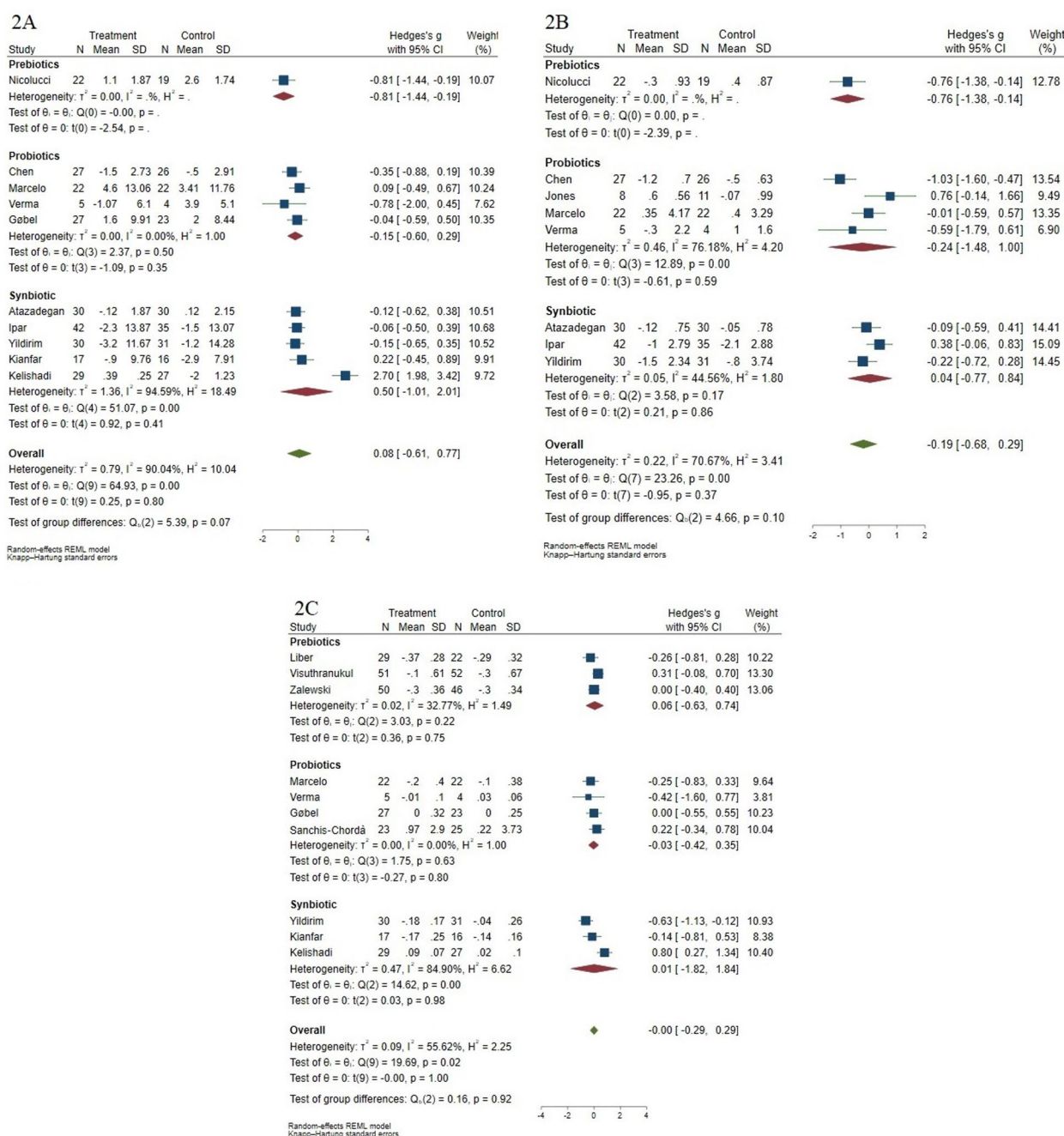


Fig. 2 Mean difference and 95% CIs presented in forest plot of the studies on the effects of biotic on weight (A), BMI (B), and BMI-Z score (C)

Biotic supplementation and glycemic indices

The finding of the meta-analysis demonstrated that prebiotic, probiotic, or synbiotics had no significant reducing effect on FBS (SMD = -0.12; 95% CI: -0.47 to 0.23; $p = 0.047$) (Fig. 3A), HOMA-IR (SMD = 0.04; 95% CI: -0.38 to 0.46; $p = 0.047$) (Fig. 3B), and insulin (SMD = 0.02; 95% CI: -0.26 to 0.30; $p = 0.047$) (Fig. 3C). Overall heterogeneity was substantial for FBS ($Q = 17.09$, $p = 0.020$; $I^2 = 60.86\%$), but not for insulin ($Q = 2.21$, $p = 0.697$; $I^2 = 0.0\%$), and HOMA-IR ($Q = 2.24$, $p = 0.523$;

$I^2 = 0.0\%$). Glycemic indices remained unchanged across all examined subgroups ($p > 0.05$). No significant effects were detected according to BMI classification, intervention duration, type of intervention, or sample size (Supplementary File).

Biotic supplementation and lipid profile

Prebiotic, probiotic, or synbiotics supplementation showed a non-significant effect on TC (SMD = -0.00; 95% CI: -0.19 to 0.19; $p = 0.374$) (Fig. 4A), TG (SMD = 0.04;

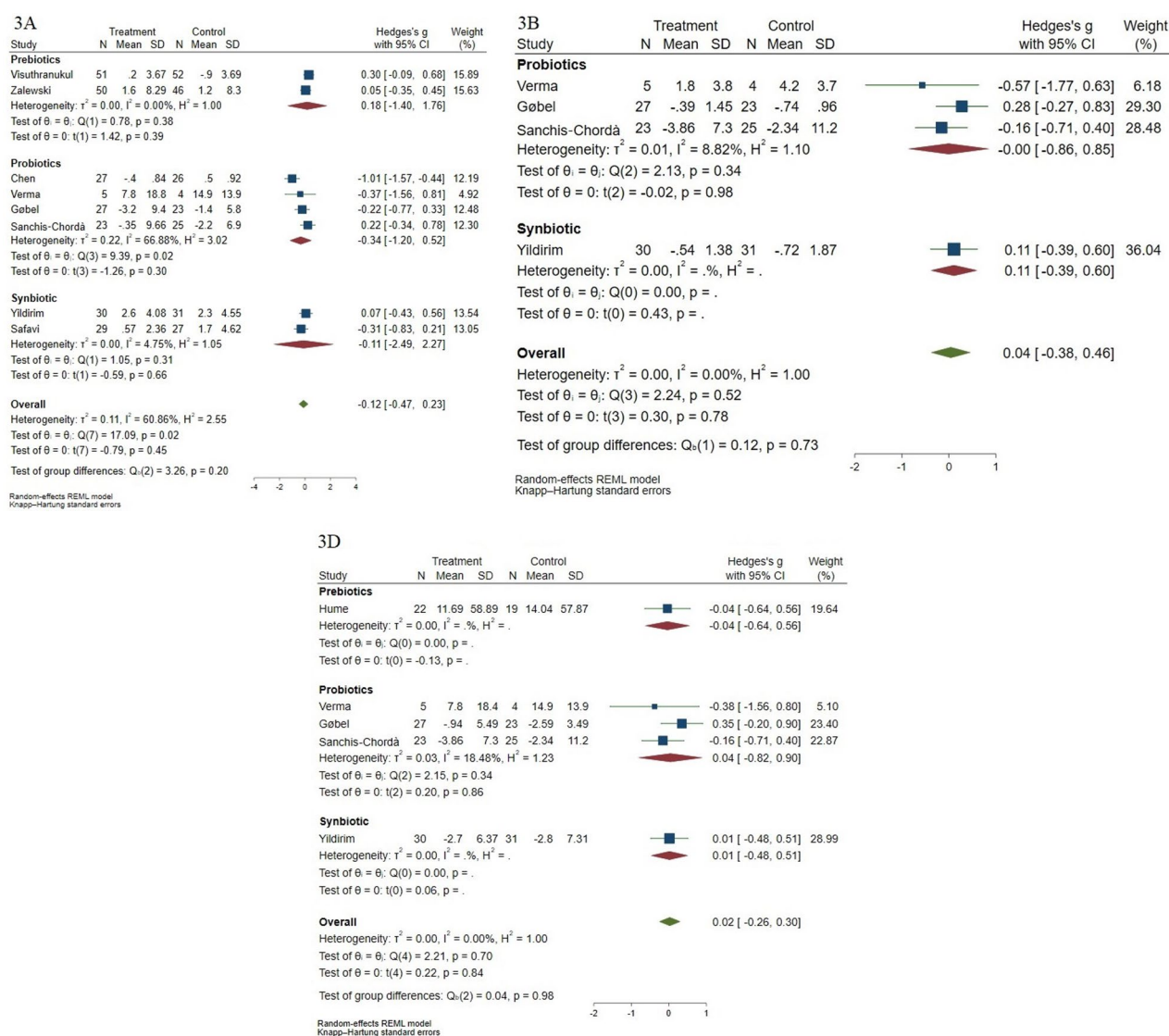


Fig. 3 Mean difference and 95% CIs presented in forest plot of the studies on the effects of biotic on FBS (A), HOMA-IR (B), and insulin (C)

95% CI: -0.22 to 0.30 ; $p = 0.374$) (Fig. 4B) and LDL-C (SMD = 0.00 ; 95% CI: -0.26 to 0.27 ; $p = 0.532$) (Fig. 4C). The pooled analysis demonstrated that prebiotic, probiotic, or synbiotics supplementation did not significantly effect on HDL-C (SMD = -0.14 ; 95% CI: -0.54 to 0.26 ; $p = 0.374$) (Fig. 4D). Overall heterogeneity was substantial for HDL-C ($Q = 17.44$, $p = 0.008$; $I^2 = 67.3\%$), but low for TC ($Q = 4.66$, $p = 0.588$; $I^2 = 0.0\%$), TG ($Q = 8.11$, $p = 0.230$; $I^2 = 23.6\%$), and LDL-C ($Q = 8.35$, $p = 0.214$; $I^2 = 26.3\%$). Subgroup analysis based on BMI classification, intervention duration, type of intervention, or sample size did not show any significant effect ($p > 0.05$) (Supplementary File).

Sensitivity analysis and publication bias

Sensitivity analysis confirmed the robustness of weight, BMI, BMI-z scores, FBS, insulin, HOMA-IR, LDL-C, HDL-C, TG, and TC results. No small-study effect was found for weight, BMI, BMI-z scores, FBS, insulin, HOMA-IR, LDL-C, HDL-C TG, and TC ($p > 0.05$) by performing Egger's and Begg's tests; however, small-study effects were detected for TG after performing Egger's test ($P < 0.05$). Funnel plot analysis was conducted only for weight and BMI z-scores, as these outcomes included at least ten studies. Visual inspection suggested asymmetry, prompting the use of the trim-and-fill method. After adjustment, the results remained non-significant. For weight, four studies were imputed (SMD = 0.45 ; 95% CI: -0.08 to 0.97 ; $p > 0.05$), and for BMI z-scores, two studies were imputed (SMD = 0.10 ; 95% CI: -0.17 to 0.37 ;

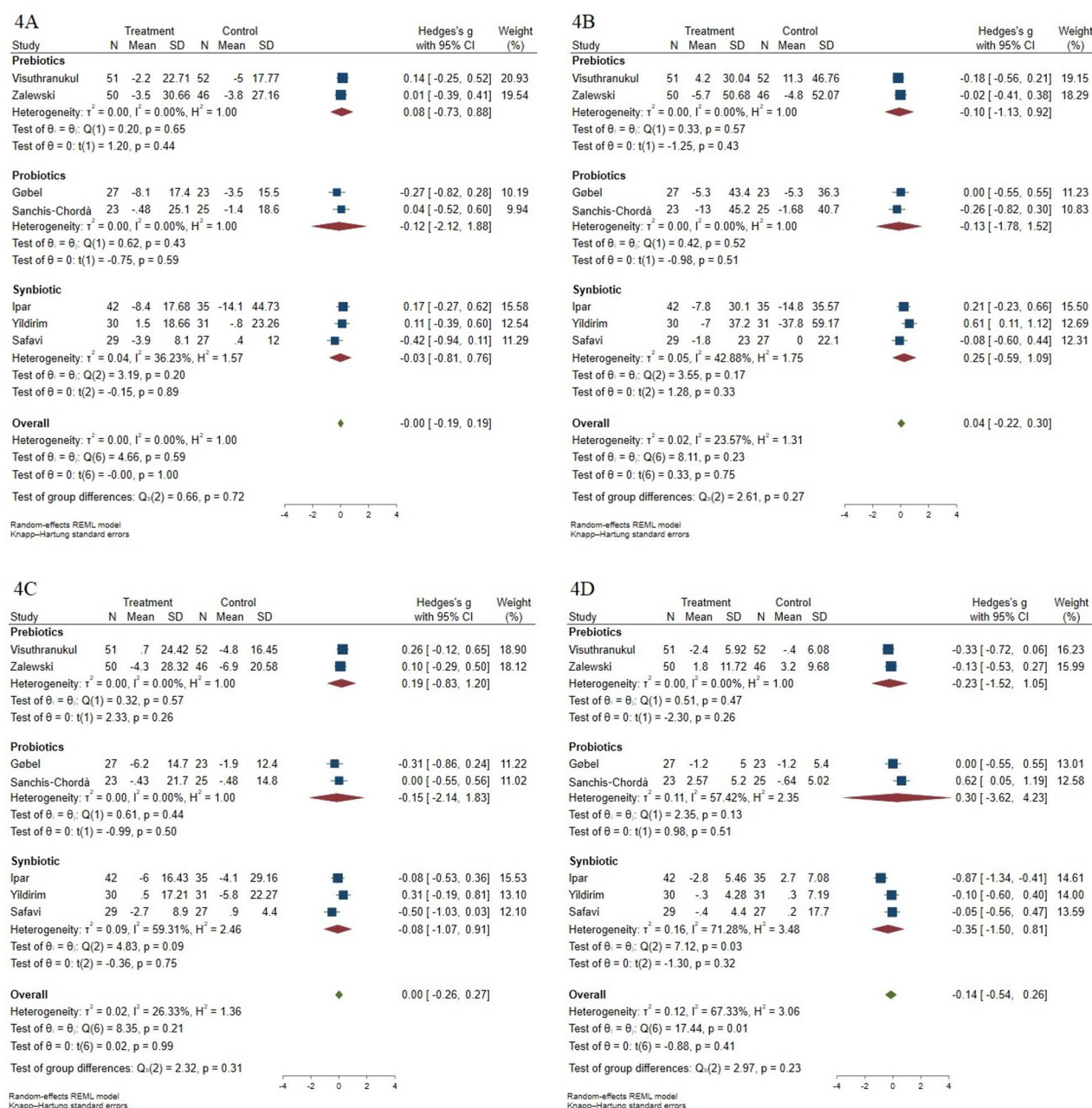


Fig. 4 Mean difference and 95% CIs presented in forest plot of the studies on the effects of biotic on TC (A), TG (B), LDL-C (C), and HDL-C (D)

$p > 0.05$). These findings indicate that potential publication bias did not materially affect the overall conclusions (Supplementary Fig. 1–2).

Discussion

In this meta-analysis of 17 randomized controlled trials ($n = 1,391$) involving overweight and obese children and adolescents, we found no significant pooled effects of probiotic, prebiotic, or synbiotic supplementation on anthropometric indices (weight, BMI, BMI z-score), glycemic outcomes (fasting glucose, insulin, HOMA-IR), or

lipid parameters (TC, LDL-C, HDL-C, TG). Subgroup findings suggested possible benefits in isolated cases. For instance, one prebiotic trial reported reductions in weight and BMI, but overall results were consistent across sensitivity analyses. Based on our results, routine supplementation with prebiotics, probiotics, or synbiotics for weight management in overweight or obese children cannot be recommended. It remains possible that specific subgroups, for example, youths with pronounced metabolic syndrome or distinct gut dysbiosis, could benefit more, but current evidence is insufficient to identify

these populations. Overall, the findings underscore the need for individualized approaches rather than universal recommendations.

Our conclusions are broadly aligned with prior pediatric reviews. Mohammadi et al. reported no significant improvements in anthropometric or metabolic outcomes with probiotics/synbiotics over 4–16 weeks [44], while Li et al. found probiotic supplementation improved lipid and adipokine profiles but did not affect BMI [45]. However, they had limited number of studies; therefore, interpretation must be performed with precaution. Duan et al. observed modest BMI and fasting glucose reductions but paradoxical increases in lipid markers [46]. Their analysis included only 8 probiotic trials (mostly from China) with approximately 390 adolescents [46], whereas our analysis included broader biotic interventions and participants. More recently, a Bayesian network meta-analysis suggested that multi-strain probiotics could reduce BMI, waist circumference, lipids, leptin, and hs-CRP, whereas synbiotics offered smaller benefits and sometimes raised fasting glucose [47]. In contrast, synbiotic supplementation was associated with modest reductions in BMI z-score and LDL-C, but also with a slight increase in fasting glucose [47]. Evidence for single-strain probiotics or prebiotics was weaker and less consistent [47]. Overall, multi-strain probiotic formulations appeared to show the greatest potential for improving obesity-related outcomes in children and adolescents [47]. The differences between our results and those of Zhang et al. (2024) [47] may be partly explained by methodological approaches. Their Bayesian network meta-analysis incorporated both direct and indirect comparisons to rank multiple interventions, which can generate broader insights but depends on assumptions of transitivity and consistency that may not always be valid in pediatric populations where strains, dosages, and dietary contexts vary considerably. By contrast, our frequentist pairwise meta-analysis relied solely on direct trial evidence, avoiding prior assumptions and yielding more conservative estimates that may better reflect actual trial data.

Findings in children and adolescents also diverge from those in adults. A large meta-analysis of more than 200 RCTs involving over 12,600 adults reported that probiotics and synbiotics led to small but statistically significant improvements, including modest weight loss (-0.91 kg), BMI reduction (-0.28 kg/m²), and decreases in waist circumference and fat mass [48]. Such differences may reflect developmental factors: children and adolescents are still undergoing growth, with unique hormonal and metabolic changes, and their gut microbiota remain in the process of maturation. These features could limit or alter their responsiveness to microbiome-targeted therapies compared with adults [49].

Mechanistically, biotics are thought to influence obesity and metabolism primarily through modulation of the gut microbiota. Prebiotics promote growth of beneficial bacteria like *Bifidobacteria* and *Lactobacilli*, leading to increased production of short-chain fatty acids (SCFAs) including acetate, propionate, and butyrate [50]. SCFAs can activate gut epithelial receptors (GPR41/43) to stimulate satiety hormones such as glucagon-like peptide-1 (GLP-1) and peptide YY (PYY), potentially reducing food intake. They may also regulate lipid metabolism and adipogenesis by upregulating peroxisome proliferator-activated receptors α and γ (PPAR α/γ) and downregulating sterol regulatory element-binding protein-1c (SREBP-1c) [51]. Probiotics can enhance gut barrier integrity, reduce endotoxin translocation, lower systemic inflammation, and improve insulin sensitivity [51]. Biotics may further influence bile acid metabolism and energy harvest.

Despite these plausible pathways, the clinical impact in pediatric populations appears limited. Longer intervention periods may be required to produce measurable anthropometric changes. Strain specificity is also important; multi-strain formulations (as noted by Zhang et al. [47]) may be more effective than single-strain products. Host factors, including baseline diet, genetics, and gut microbiota composition, likely contribute to individual variability in response. In summary, although mechanistic evidence supports potential metabolic benefits, our pooled data indicate that effects of biotics on weight, glycemia, and lipids in children and adolescents are small and inconsistent.

This study has several limitations. The included trials differed considerably in participant age, baseline obesity severity, probiotic strains, prebiotic types, dosages, and intervention durations (ranging from 4 to 24 weeks), which contributed to substantial statistical heterogeneity (e.g., $I^2 > 80\%$ for weight), making pooled estimates less precise. Second, some outcomes, such as HOMA-IR and insulin, were reported in only a few studies, resulting in limited evidence for these measures. Third, since most of the included studies did not report results separately for children and adolescents, we were unable to perform subgroup analyses based on age groups. Fourth, low heterogeneity in FBS, HOMA-IR, insulin, TC, LDL-c, and TG values suggests consistency across the included studies. However, this should be interpreted with caution, as the number of studies for these outcomes was limited.

The strengths of this study include the comprehensive inclusion of all available RCTs evaluating prebiotics, probiotics, and synbiotics in children and adolescents with obesity. To our knowledge, this represents the largest pediatric-focused meta-analysis on this topic. We investigated potential sources of heterogeneity by conducting stratified analyses based on intervention type, study duration, and sample size. The review adhered to PRISMA

guidelines, and the quality of evidence was assessed using the GRADE framework. Additionally, sensitivity analyses (leave-one-out) and tests for publication bias were performed across all outcomes to ensure the robustness of our findings.

Conclusion

Biotic supplementation has not been shown to consistently improve metabolic syndrome risk factors in overweight and obese children, except for a modest beneficial effect of prebiotics on weight and BMI (with very-low certainty of evidence); however, alternative probiotic organisms or formulations not tested to date may have different effects. Overall, these findings do not support the routine use of biotics for pediatric weight management, and future research should focus on longer-duration trials, specific subgroups with metabolic dysregulation, and standardized intervention protocols to clarify potential benefits.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

Jiamin Xie: Conceptualization, study design, supervision, methodology, data interpretation, manuscript drafting, and corresponding author responsibilities. Sho Liu: Literature search, data extraction, formal analysis, visualization, and writing – original draft. Xiao Wong: Data curation, quality assessment, validation of results, and manuscript review and editing. All authors reviewed the manuscript.

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Data availability

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

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Competing interests

The authors declare no competing interests.

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